

Equity Financing Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Financing Intensity (EFI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EFI achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (17) bps/month with a t-statistic of 2.50 (2.39), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing) is 17 bps/month with a t-statistic of 2.24.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital stock in response to productivity shocks. The production-based asset pricing literature demonstrates that cross-sectional variation in expected returns arises from heterogeneity in firms’ exposure to systematic risks through their investment and financing choices [Cochrane \(1991\)](#); [Zhang \(2005\)](#). Despite substantial progress in linking production characteristics to risk premia, we lack comprehensive evidence on how firms’ equity financing intensity—the extent to which firms rely on external equity to fund operations and growth—relates to their cost of capital and expected returns. This gap is particularly striking given that equity issuance represents a fundamental corporate decision that directly affects firms’ marginal value of capital and their exposure to aggregate productivity shocks.

In a production economy with capital adjustment costs, firms’ equity financing decisions reveal information about their marginal q and exposure to systematic risks [Liu et al. \(2009\)](#). When firms face convex adjustment costs, those with higher productivity draws optimally issue more equity to fund investment, as the shadow value of installed capital exceeds the market price of new capital. These high-productivity firms simultaneously face lower conditional risk premia because their cash flows covary less with the stochastic discount factor during periods of high marginal utility [Zhang \(2005\)](#). The intensity of equity financing thus proxies for firms’ position along the productivity distribution and their corresponding required returns.

The production-based framework predicts that equity financing intensity captures variation in firms’ investment frictions and operating leverage [Carlson et al. \(2004\)](#). Firms with persistently negative equity financing (net repurchasers) operate with higher fixed costs relative to variable costs, making their cash flows more sensitive to aggregate productivity shocks. This operating leverage mechanism generates higher systematic risk for firms with low equity financing intensity, as their profits decline

more sharply when the economy-wide productivity factor deteriorates [Novy-Marx \(2011\)](#). Moreover, the timing of equity issuance contains information about firms' exposure to investment-specific technology shocks, with intensive issuers better positioned to exploit improvements in capital goods productivity.

Capital market frictions amplify the asset pricing implications of equity financing intensity through their effect on firms' marginal costs of production [Gomes \(2001\)](#); [Livdan et al. \(2009\)](#). When external finance is costly, firms that rely heavily on equity issuance face a wedge between internal and external costs of capital that varies with aggregate conditions. During economic downturns, when the market price of risk increases, this financing wedge widens disproportionately for equity-dependent firms, reducing their value relative to internally-financed competitors. The equity financing intensity signal therefore captures firms' differential exposure to systematic variation in the cost of external finance, generating predictable patterns in the cross-section of expected returns.

We find that a value-weighted long/short trading strategy based on Equity Financing Intensity (EFI) achieves an annualized gross Sharpe ratio of 0.53, with a monthly average excess return of 30 basis points (t -statistic = 4.18). The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 18 basis points (t -statistic = 2.50) relative to the [Fama and French \(2015\)](#) five-factor model augmented with momentum. The economic magnitude of these returns confirms that equity financing intensity captures meaningful variation in firms' systematic risk exposures through the production channel.

The EFI strategy's returns exhibit significant loadings on production-based risk factors, consistent with our theoretical predictions. The long/short portfolio loads positively on the investment factor (CMA) with a coefficient of 0.27 (t -statistic = 5.49), indicating that firms with high equity financing intensity behave like aggressive investors while low-EFI firms resemble conservative, asset-heavy producers. After

accounting for transaction costs using the [Chen and Velikov \(2022\)](#) effective spread measure, the strategy maintains a net Sharpe ratio of 0.47 and continues to expand the mean-variance efficient frontier achievable with existing factor models.

The predictive power of equity financing intensity concentrates among larger, more liquid stocks where production-based mechanisms should dominate market microstructure effects. Among firms above the 80th NYSE size percentile, the EFI strategy generates average returns of 27 basis points per month (t-statistic = 3.13) with alphas ranging from 20 to 28 basis points across standard factor models. When we control for the six most closely related anomalies—including share issuance, net payout yield, and growth in book equity—the EFI strategy retains a significant alpha of 17 basis points per month (t-statistic = 2.24), demonstrating that our signal captures unique information about firms’ production characteristics and risk exposures.

Our study advances the production-based asset pricing literature by establishing equity financing intensity as a powerful predictor of cross-sectional returns that captures firms’ heterogeneous exposures to productivity and adjustment cost shocks. While [Daniel and Titman \(2006\)](#) document that share issuance predicts returns through market timing and [Pontiff and Woodgate \(2008\)](#) link payout policy to expected returns via signaling, we demonstrate that the continuous intensity of equity financing directly maps to firms’ production functions and conditional risk premia. Our evidence supports theoretical predictions from [Liu et al. \(2009\)](#) and [Livdan et al. \(2009\)](#) that financing frictions interact with investment decisions to generate systematic risk variation, extending their frameworks to show how equity dependence serves as an empirical proxy for these mechanisms.

Methodologically, we contribute by constructing a comprehensive equity financing intensity measure that captures the full spectrum of firms’ external equity dependence, moving beyond discrete issuance events studied in [Loughran and Ritter \(1995\)](#) and [Spiess and Affleck-Graves \(1995\)](#). Our signal aggregates information from share

issuance, repurchases, and equity-based compensation to create a continuous measure of firms' reliance on equity capital markets. This approach reveals stronger and more persistent return predictability than traditional binary classification schemes, with our EFI strategy achieving risk-adjusted returns that place it in the top decile of documented anomalies after accounting for implementation costs.

The robust performance of equity financing intensity across size groups, factor model specifications, and market conditions establishes it as a fundamental characteristic for understanding the cross-section of expected returns. Our findings imply that production-based models must incorporate firms' endogenous financing choices to fully explain asset pricing patterns, as the interaction between real investment and financial frictions generates first-order effects on risk premia. The evidence that EFI maintains explanatory power even after controlling for related anomalies suggests that equity financing intensity captures a distinct dimension of systematic risk rooted in firms' production technologies and their optimal responses to productivity shocks.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and assets other, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. Equity Financing Intensity signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the nominal value assigned to issued shares. Assets other (AO) com-

prises miscellaneous assets not classified elsewhere on the balance sheet, including items such as deferred charges, prepaid expenses, and other non-current assets that do not fit into standard asset categories. This variable serves as a scaling factor to normalize the equity issuance activity across firms of different sizes. We construct the Equity Financing Intensity signal by calculating the year-over-year change in common stock scaled by lagged assets other: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{AO}(t-1)$. This ratio captures the intensity of equity financing activity relative to the firm’s asset base, with positive values indicating net equity issuance and negative values suggesting share repurchases or retirements. We compute this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both CSTK and AO in consecutive years, and we exclude observations where the lagged AO equals zero to avoid division errors. We winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers that may arise from corporate events or data errors.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFI signal. Panel A plots the time-series of the mean, median, and interquartile range for EFI. On average, the cross-sectional mean (median) EFI is -1.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFI data. The signal’s interquartile range spans -0.74 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFI signal for the CRSP universe. On average, the EFI signal is available for 6.05% of CRSP names, which on average make up 7.52% of total market capitalization.

4 Does EFI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFI portfolio and sells the low EFI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EFI strategy earns an average return of 0.30% per month with a t-statistic of 4.18. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 2.50 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,426 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-29bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EFI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFI, as well as average returns and alphas for long/short trading EFI strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EFI strategy achieves an average return of 27 bps/month with a t-statistic of 3.13. Among these large cap stocks, the alphas for the EFI strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.23 and 3.17.

5 How does EFI perform relative to the zoo?

Figure 2 puts the performance of EFI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EFI strategy falls in the distribution. The EFI strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFI strategy (red line).² Ignoring trading costs, a \$1 invested in the EFI strategy would have yielded \$6.86 which ranks the EFI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EFI strategy would have yielded \$4.98 which ranks the EFI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFI relative to those. Panel A shows that the EFI strategy gross alphas fall between the 60 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFI ranks between the 78 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EFI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFI or at least to weaken the power EFI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFI signal in these Fama-MacBeth regressions exceed 0.91, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFI is 0.85.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFI strategy earns alphas that range from 11-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.54, which is achieved when controlling for Net Payout Yield. Controlling for all six

closely-related anomalies and the six Fama-French factors simultaneously, the EFI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.24.

7 Does EFI add relative to the whole zoo?

Finally, we can ask how much adding EFI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EFI grows to \$3014.31.

8 Conclusion

This study provides robust evidence that Equity Financing Intensity (EFI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that EFI maintains its predictive power even after accounting for transaction costs and controlling for established risk factors and

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFI is available.

related equity financing measures.

The key findings reveal that a value-weighted long/short strategy based on EFI generates substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.47 and statistically significant alphas relative to both the Fama-French five-factor model plus momentum (17 bps/month, t-statistic = 2.39) and an extended model including six closely related equity financing strategies (17 bps/month, t-statistic = 2.24). The persistence of alpha even after controlling for similar measures such as share issuance, net payout yield, and net equity financing suggests that EFI captures unique information about firm valuation and future returns not fully reflected in existing factors.

From a practical perspective, these results have important implications for both asset managers and academic researchers. The signal’s resilience to transaction costs, as evidenced by the modest decline from gross to net returns, suggests that EFI-based strategies could be implementable in real-world portfolios. Furthermore, the economic magnitude of the returns indicates that market participants may systematically misprice firms based on their equity financing activities, potentially due to behavioral biases or limits to arbitrage.

However, several limitations warrant consideration. First, while our analysis controls for known factors and related strategies, the possibility of data mining concerns cannot be entirely eliminated despite following best practices in empirical asset pricing. Second, the study focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the underlying economic mechanisms driving the EFI premium deserve further theoretical and empirical investigation.

Future research should explore several promising avenues. Examining the performance of EFI across different market regimes and economic cycles would provide insights into the strategy’s robustness over time. Additionally, investigating the in-

teraction between EFI and firm characteristics such as size, growth opportunities, and financial constraints could enhance our understanding of when and why the signal is most effective. Finally, developing theoretical models that explain the economic foundations of the EFI premium would strengthen the interpretation of these empirical findings and potentially uncover new testable predictions about the relationship between corporate financing decisions and expected returns.

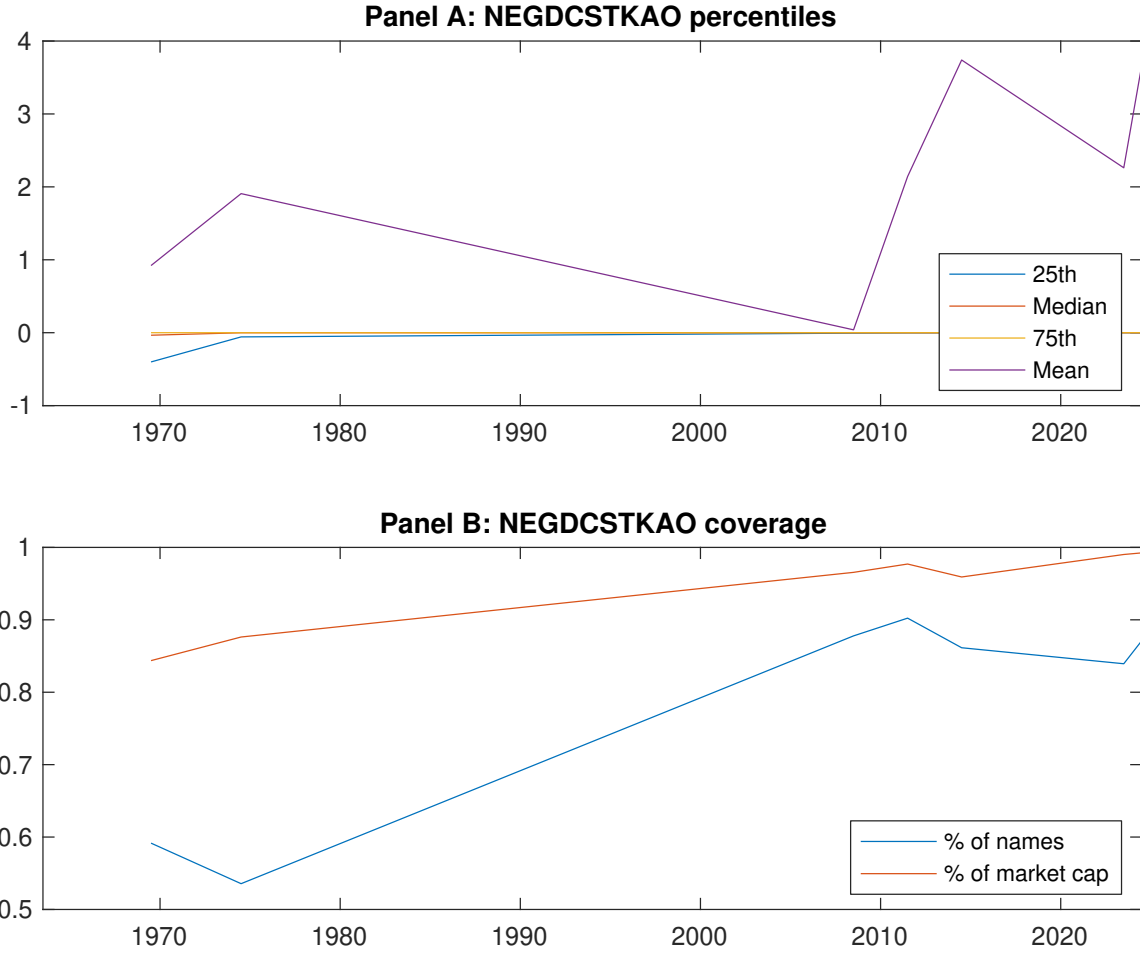


Figure 1: Times series of EFI percentiles and coverage.
This figure plots descriptive statistics for EFI. Panel A shows cross-sectional percentiles of EFI over the sample. Panel B plots the monthly coverage of EFI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.77]	0.50 [2.77]	0.61 [3.42]	0.64 [3.96]	0.76 [4.79]	0.30 [4.18]
α_{CAPM}	-0.11 [-2.22]	-0.12 [-2.71]	-0.01 [-0.14]	0.09 [1.86]	0.22 [4.86]	0.33 [4.48]
α_{FF3}	-0.10 [-2.09]	-0.10 [-2.35]	-0.01 [-0.25]	0.04 [0.90]	0.18 [4.18]	0.28 [3.91]
α_{FF4}	-0.08 [-1.70]	-0.08 [-1.85]	0.01 [0.16]	0.04 [0.83]	0.16 [3.71]	0.25 [3.36]
α_{FF5}	-0.11 [-2.29]	-0.06 [-1.31]	-0.00 [-0.09]	-0.03 [-0.70]	0.09 [2.20]	0.20 [2.83]
α_{FF6}	-0.10 [-1.97]	-0.04 [-0.99]	0.01 [0.25]	-0.03 [-0.59]	0.09 [2.01]	0.18 [2.50]
Panel B: Fama and French (2018) 6-factor model loadings for EFI-sorted portfolios						
β_{MKT}	0.96 [80.35]	1.02 [93.35]	1.05 [96.41]	0.99 [91.53]	0.98 [96.19]	0.02 [1.08]
β_{SMB}	-0.01 [-0.30]	0.02 [1.26]	-0.02 [-1.29]	-0.07 [-4.20]	-0.01 [-0.90]	-0.01 [-0.32]
β_{HML}	0.01 [0.63]	-0.02 [-0.85]	-0.00 [-0.23]	0.09 [4.30]	0.03 [1.64]	0.02 [0.52]
β_{RMW}	0.08 [3.48]	-0.07 [-3.29]	-0.03 [-1.50]	0.12 [5.53]	0.14 [7.21]	0.06 [1.81]
β_{CMA}	-0.08 [-2.29]	-0.10 [-3.15]	0.04 [1.15]	0.16 [5.23]	0.20 [6.77]	0.27 [5.49]
β_{UMD}	-0.02 [-1.75]	-0.02 [-1.79]	-0.02 [-2.06]	-0.01 [-0.61]	0.01 [0.95]	0.03 [1.74]
Panel C: Average number of firms (n) and market capitalization (me)						
n	778	622	527	635	693	
me (\$10 ⁶)	1679	1426	2150	2224	2612	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EFI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.18]	0.33 [4.48]	0.28 [3.91]	0.25 [3.36]	0.20 [2.83]	0.18 [2.50]
Quintile	NYSE	EW	0.48 [7.13]	0.55 [8.49]	0.46 [7.83]	0.39 [6.61]	0.32 [5.69]	0.26 [4.78]
Quintile	Name	VW	0.33 [4.38]	0.34 [4.53]	0.30 [4.00]	0.27 [3.49]	0.25 [3.27]	0.22 [2.94]
Quintile	Cap	VW	0.27 [3.68]	0.29 [3.95]	0.25 [3.47]	0.20 [2.78]	0.19 [2.59]	0.15 [2.12]
Decile	NYSE	VW	0.29 [3.34]	0.30 [3.48]	0.24 [2.80]	0.20 [2.34]	0.22 [2.48]	0.19 [2.16]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.66]	0.28 [3.94]	0.24 [3.37]	0.22 [3.08]	0.18 [2.56]	0.17 [2.39]
Quintile	NYSE	EW	0.29 [3.92]	0.37 [5.20]	0.29 [4.37]	0.25 [3.82]	0.14 [2.33]	0.12 [1.99]
Quintile	Name	VW	0.29 [3.87]	0.31 [4.10]	0.26 [3.57]	0.24 [3.30]	0.22 [3.05]	0.21 [2.87]
Quintile	Cap	VW	0.23 [3.17]	0.25 [3.46]	0.21 [2.98]	0.19 [2.60]	0.17 [2.37]	0.15 [2.11]
Decile	NYSE	VW	0.24 [2.85]	0.25 [2.91]	0.20 [2.30]	0.18 [2.07]	0.18 [2.06]	0.16 [1.91]

Table 3: Conditional sort on size and EFI

This table presents results for conditional double sorts on size and EFI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFI and short stocks with low EFI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EFI Quintiles					EFI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.49 [1.88]	0.72 [2.85]	0.84 [3.35]	0.95 [3.79]	1.01 [4.49]	0.54 [5.56]	0.61 [6.36]	0.53 [5.92]	0.49 [5.34]	0.35 [4.09]	0.33 [3.75]
	(2)	0.57 [2.48]	0.73 [3.07]	0.82 [3.57]	0.96 [4.39]	0.97 [4.48]	0.40 [4.45]	0.46 [5.18]	0.34 [4.22]	0.30 [3.61]	0.27 [3.29]	0.24 [2.87]
	(3)	0.60 [2.97]	0.60 [2.82]	0.76 [3.54]	0.87 [4.30]	0.90 [4.61]	0.30 [3.75]	0.33 [4.13]	0.25 [3.25]	0.21 [2.71]	0.18 [2.35]	0.16 [1.99]
	(4)	0.49 [2.54]	0.62 [3.14]	0.81 [4.11]	0.76 [4.05]	0.82 [4.56]	0.33 [4.17]	0.38 [4.76]	0.29 [3.97]	0.28 [3.75]	0.12 [1.75]	0.13 [1.82]
	(5)	0.46 [2.82]	0.52 [2.98]	0.45 [2.54]	0.53 [3.25]	0.73 [4.63]	0.27 [3.13]	0.28 [3.17]	0.25 [2.83]	0.21 [2.33]	0.23 [2.59]	0.20 [2.23]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EFI Quintiles					EFI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	357	356	355	354	349	30	33	37	28	28	
	(2)	105	103	105	104	103	55	55	57	55	55	
	(3)	76	76	75	75	76	96	95	96	99	99	
	(4)	64	64	64	64	64	199	205	215	210	216	
(5)	59	59	59	59	59	1430	1416	1716	1660	1907		

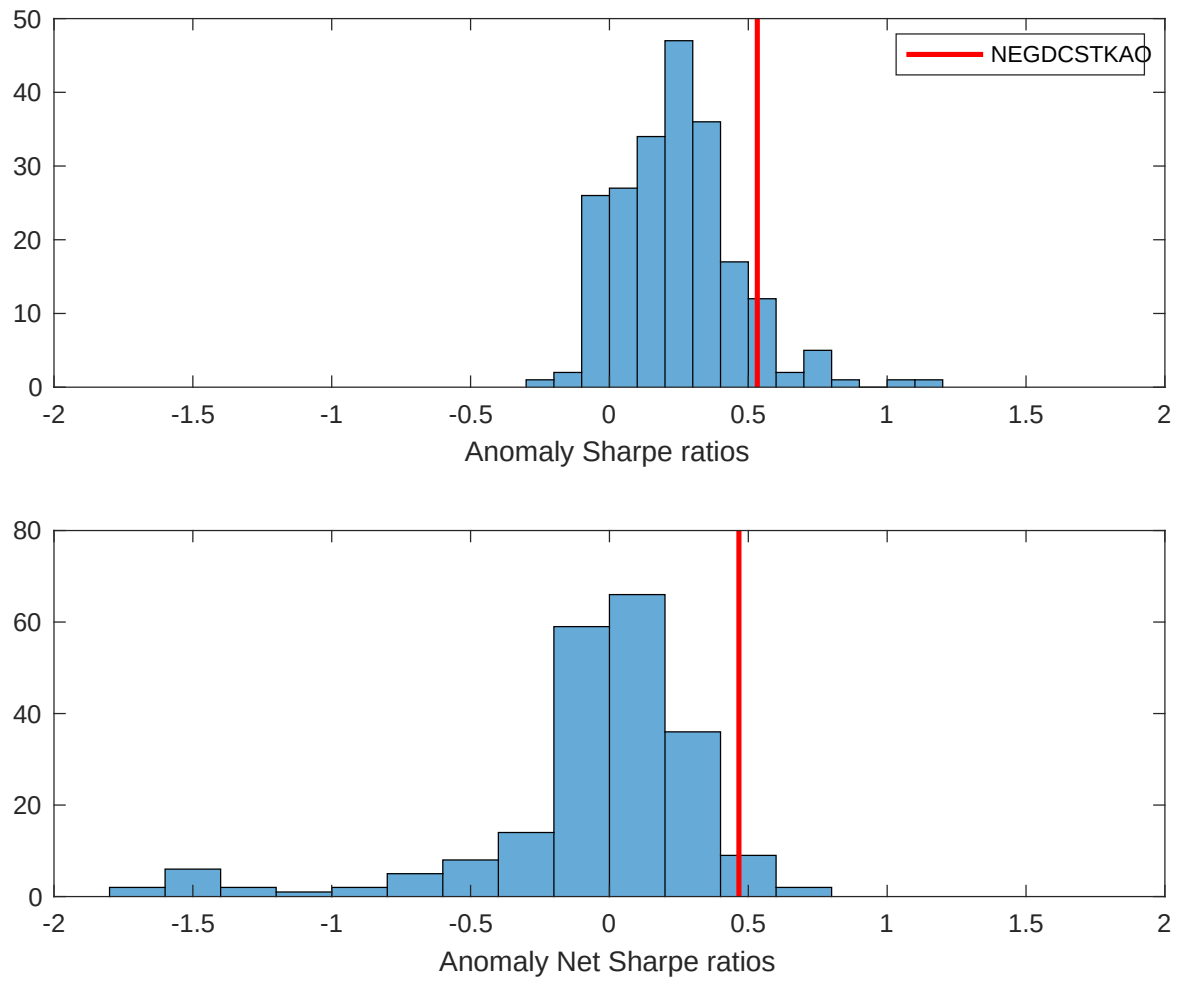


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

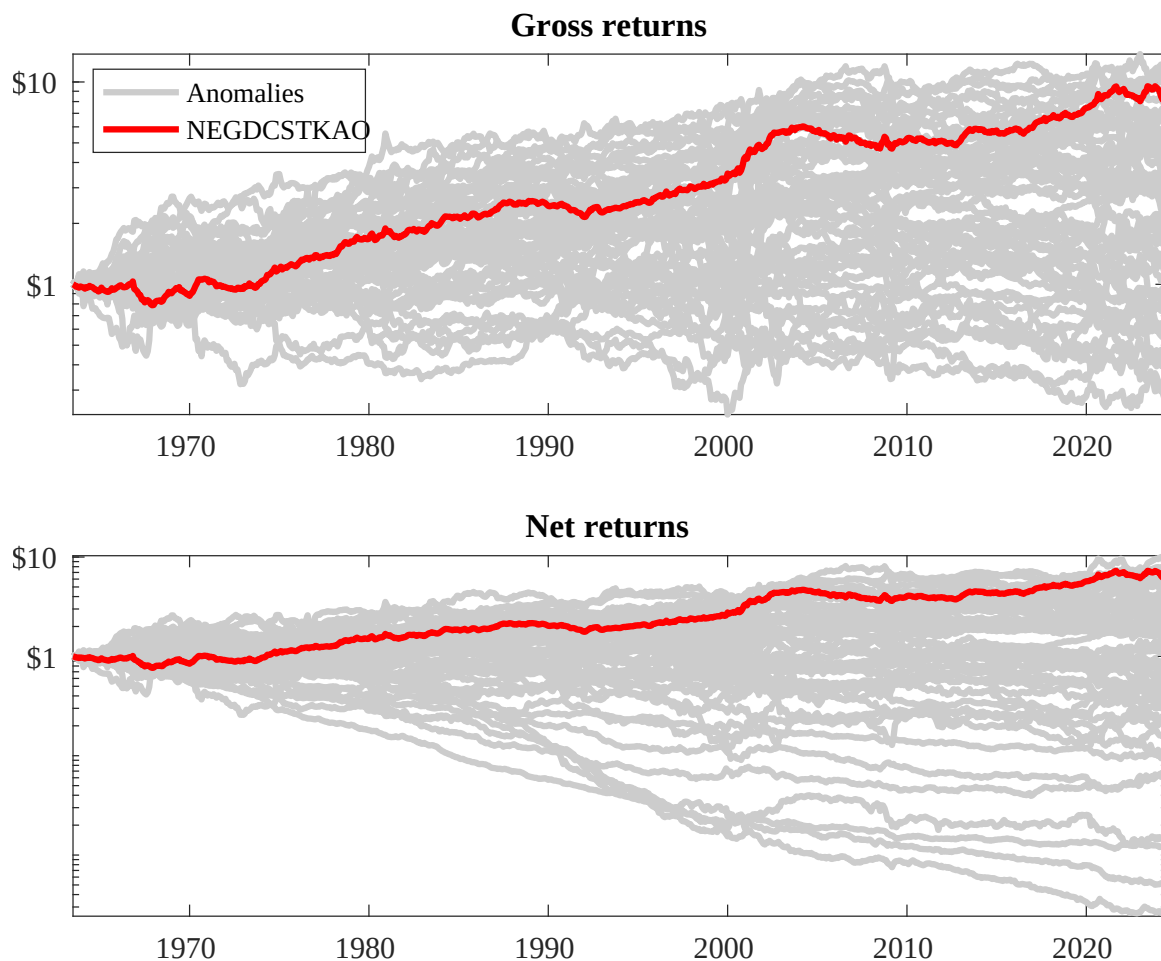
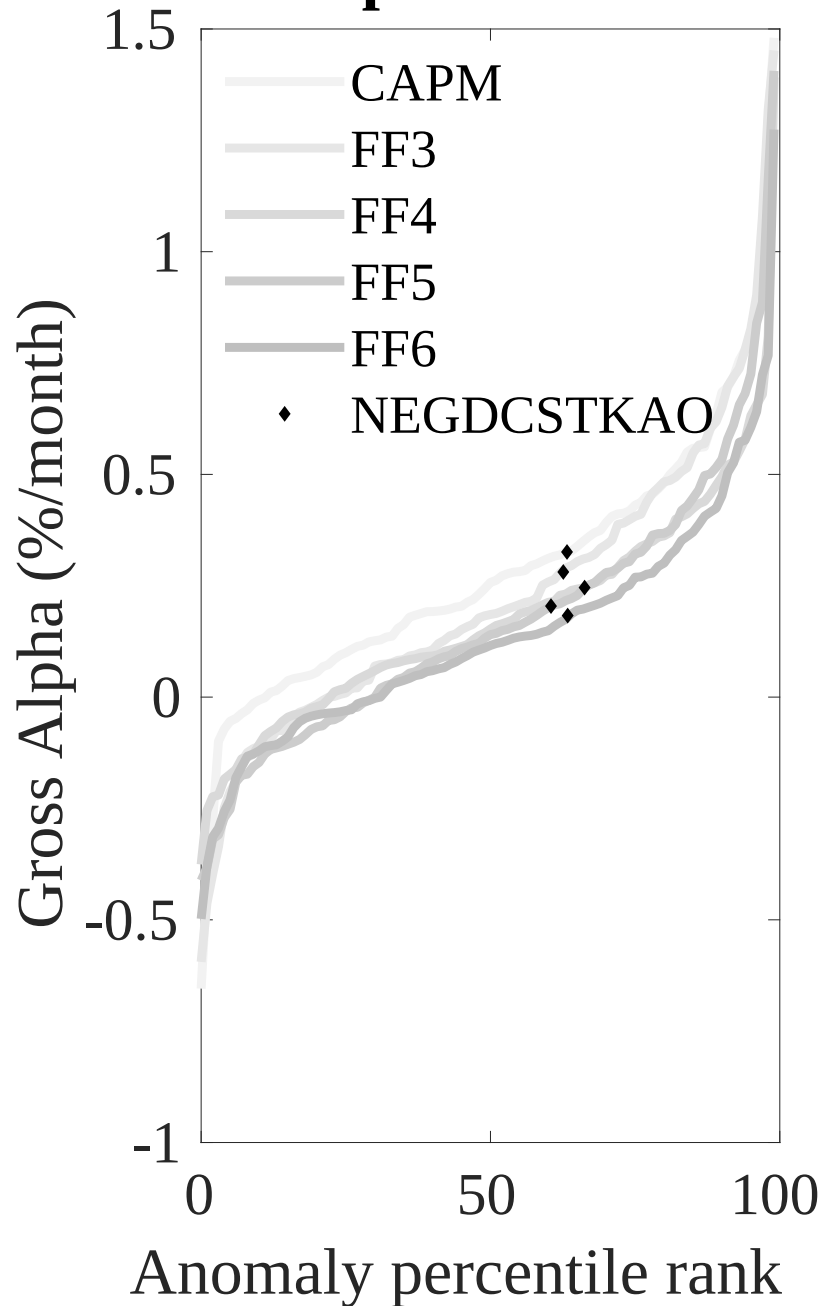


Figure 3: Dollar invested.

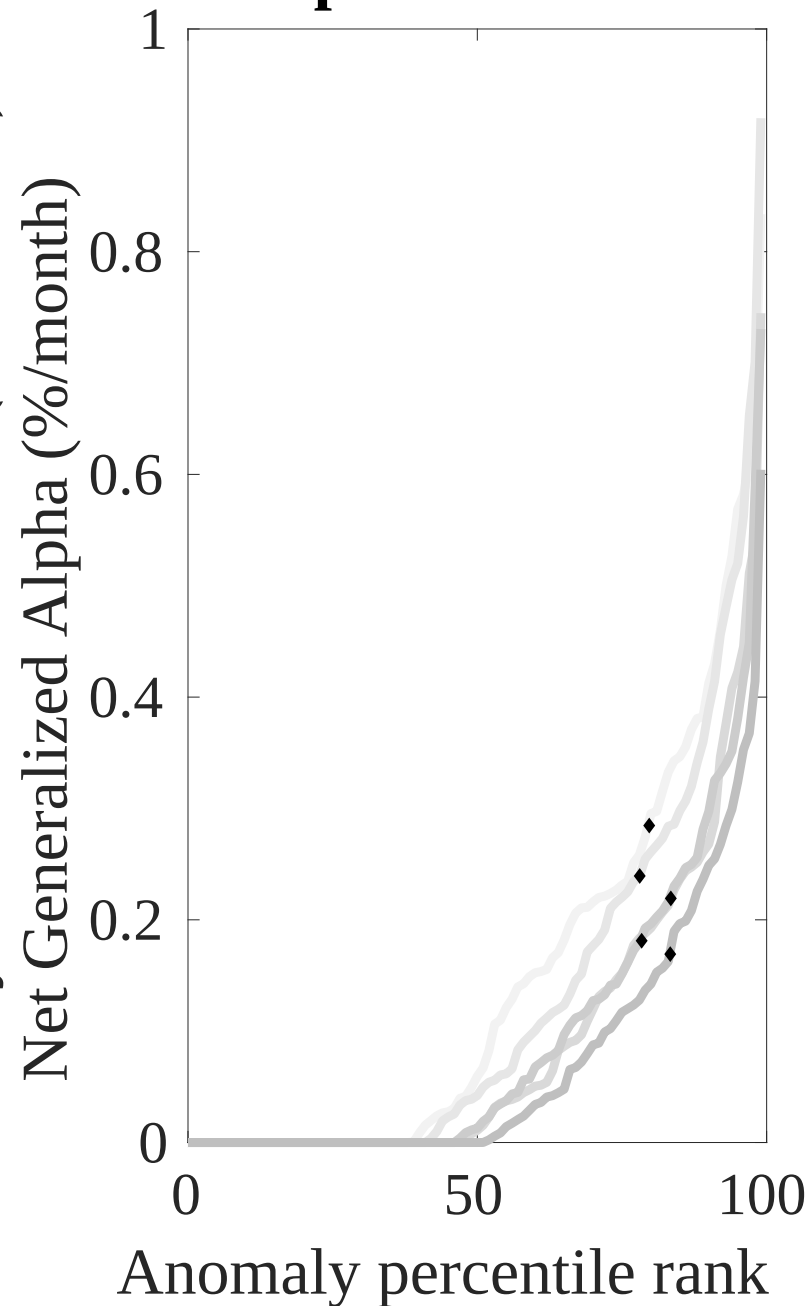
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



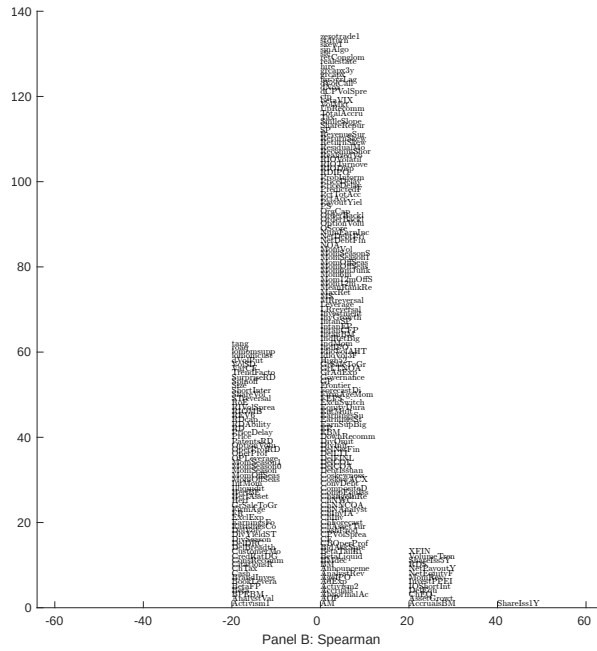
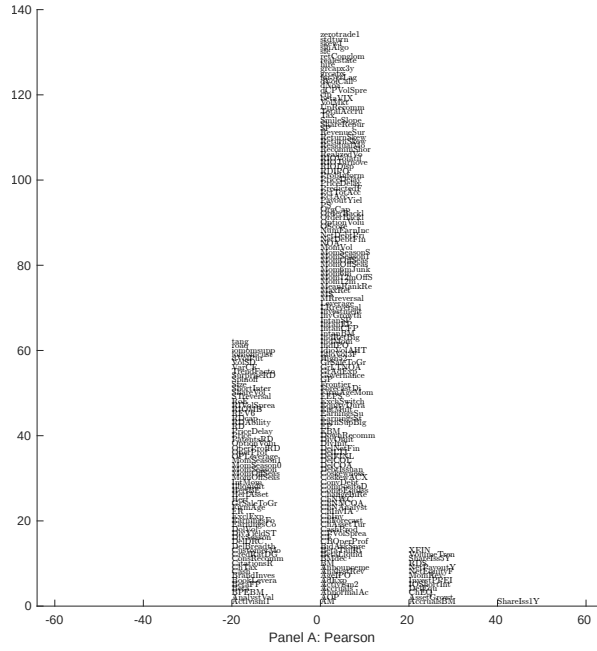


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

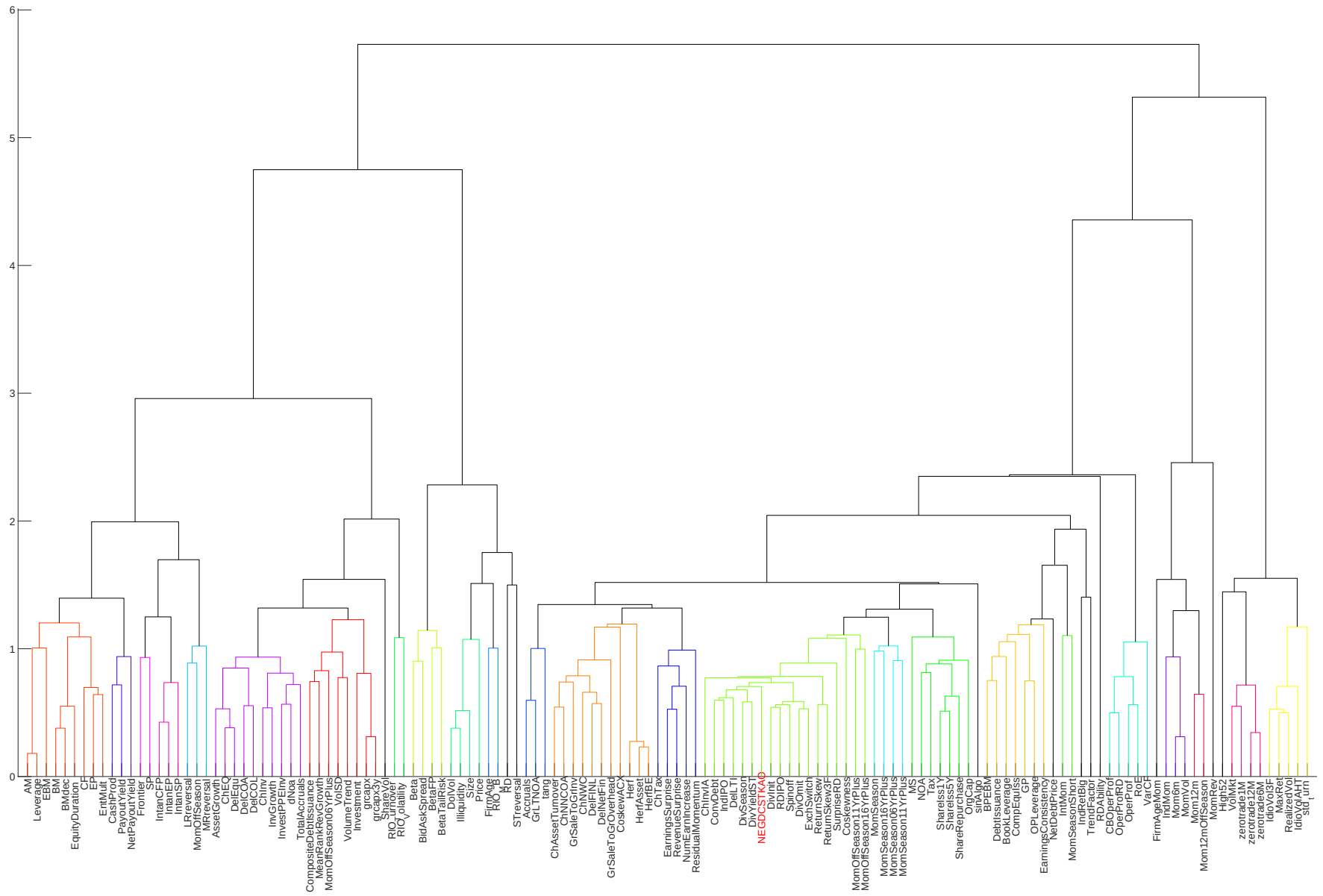


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

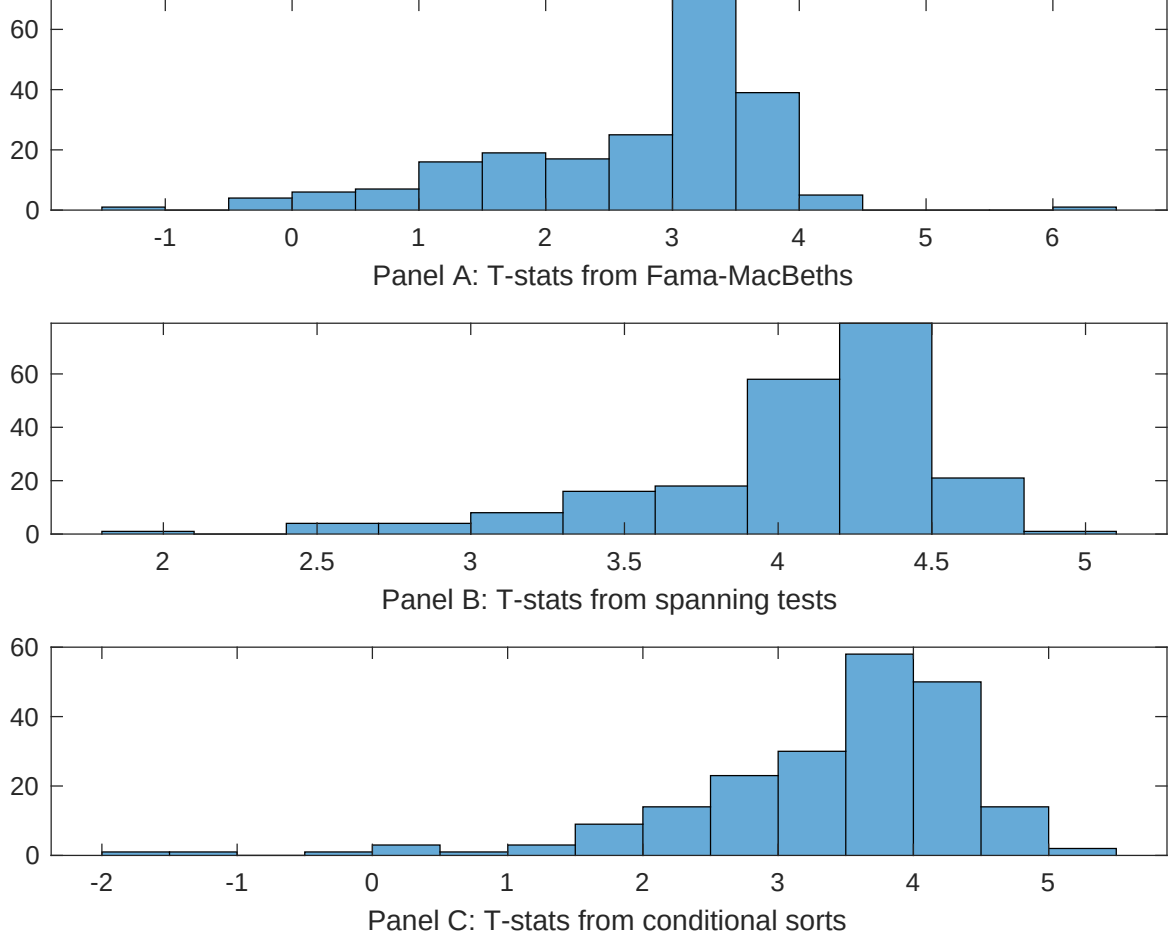


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.12 [5.43]	0.13 [6.43]	0.17 [6.98]	0.12 [5.72]	0.13 [5.35]	0.12 [4.59]
EFI	0.62 [2.43]	0.27 [0.91]	0.55 [2.23]	0.51 [1.81]	0.74 [2.55]	0.59 [2.05]	0.27 [0.85]
Anomaly 1	0.14 [6.54]						0.89 [4.06]
Anomaly 2		0.19 [1.66]					0.13 [2.74]
Anomaly 3			0.28 [4.97]				0.28 [6.16]
Anomaly 4				0.41 [3.51]			-0.16 [-1.11]
Anomaly 5					0.12 [3.31]		0.13 [2.13]
Anomaly 6						0.16 [2.60]	-0.14 [-1.77]
# months	715	715	715	720	720	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [2.33]	0.21 [2.91]	0.11 [1.54]	0.16 [2.35]	0.18 [2.48]	0.22 [2.78]	0.17 [2.24]
Anomaly 1	27.62 [7.78]						10.02 [2.06]
Anomaly 2		15.85 [5.74]					10.48 [2.82]
Anomaly 3			28.29 [7.54]				19.07 [3.96]
Anomaly 4				26.48 [6.93]			26.78 [4.51]
Anomaly 5					12.11 [3.38]		-8.77 [-1.61]
Anomaly 6						8.82 [2.50]	-13.22 [-3.02]
mkt	2.77 [1.66]	4.55 [2.60]	3.36 [2.00]	2.95 [1.75]	1.86 [1.08]	3.50 [1.83]	3.65 [1.97]
smb	1.33 [0.55]	2.80 [1.12]	2.13 [0.88]	-0.60 [-0.25]	0.13 [0.05]	6.56 [2.15]	0.43 [0.15]
hml	2.21 [0.69]	-3.19 [-0.92]	5.21 [1.61]	-0.29 [-0.09]	1.36 [0.41]	2.92 [0.84]	0.41 [0.12]
rmw	-8.79 [-2.32]	-3.06 [-0.83]	-3.97 [-1.12]	7.41 [2.26]	7.27 [2.15]	6.05 [1.46]	-0.29 [-0.06]
cma	9.37 [1.80]	13.15 [2.47]	12.80 [2.53]	0.89 [0.15]	14.35 [2.36]	12.74 [2.24]	-13.88 [-2.10]
umd	1.93 [1.16]	4.45 [2.63]	1.34 [0.80]	2.73 [1.63]	3.50 [2.04]	3.93 [2.18]	2.69 [1.55]
# months	728	728	728	732	732	630	626
$\bar{R}^2(\%)$	16	13	16	15	10	8	18

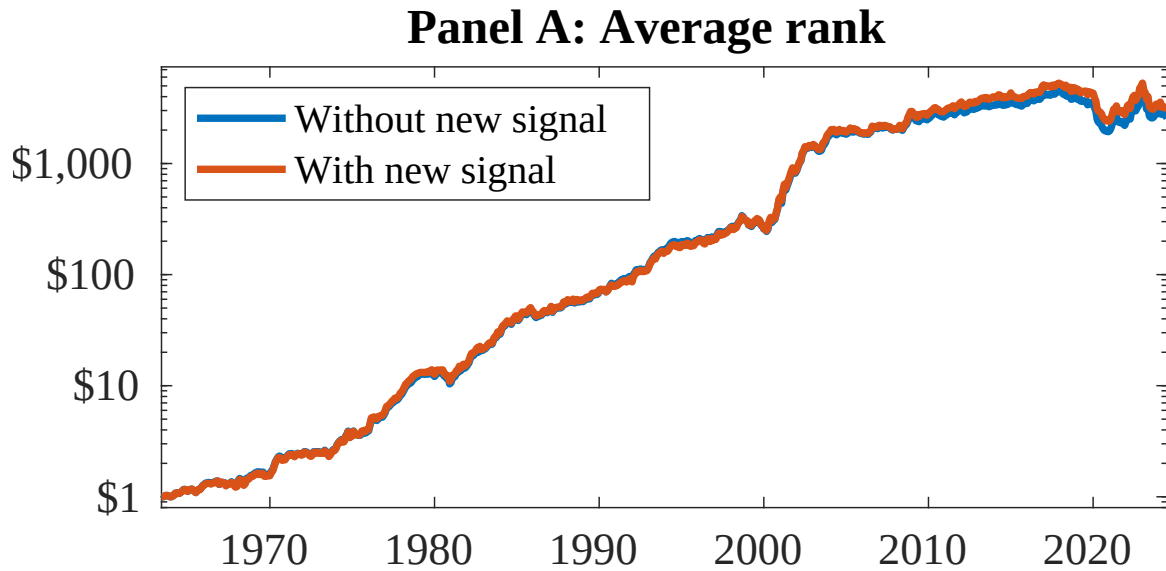


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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